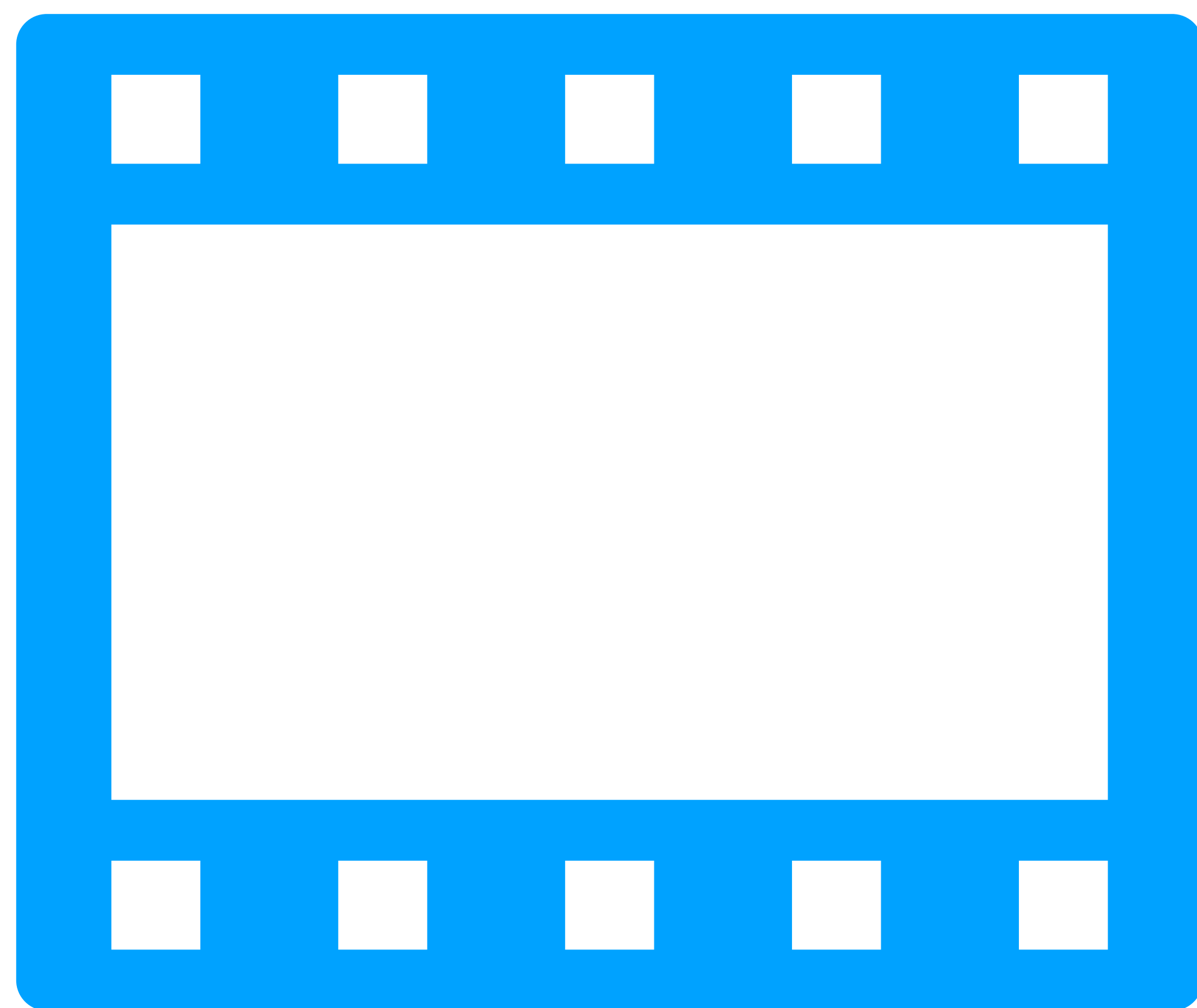


Attack Modelling

Dr. Ramokapane



Why model attacks?

A formalised way of understanding how systems can be attacked

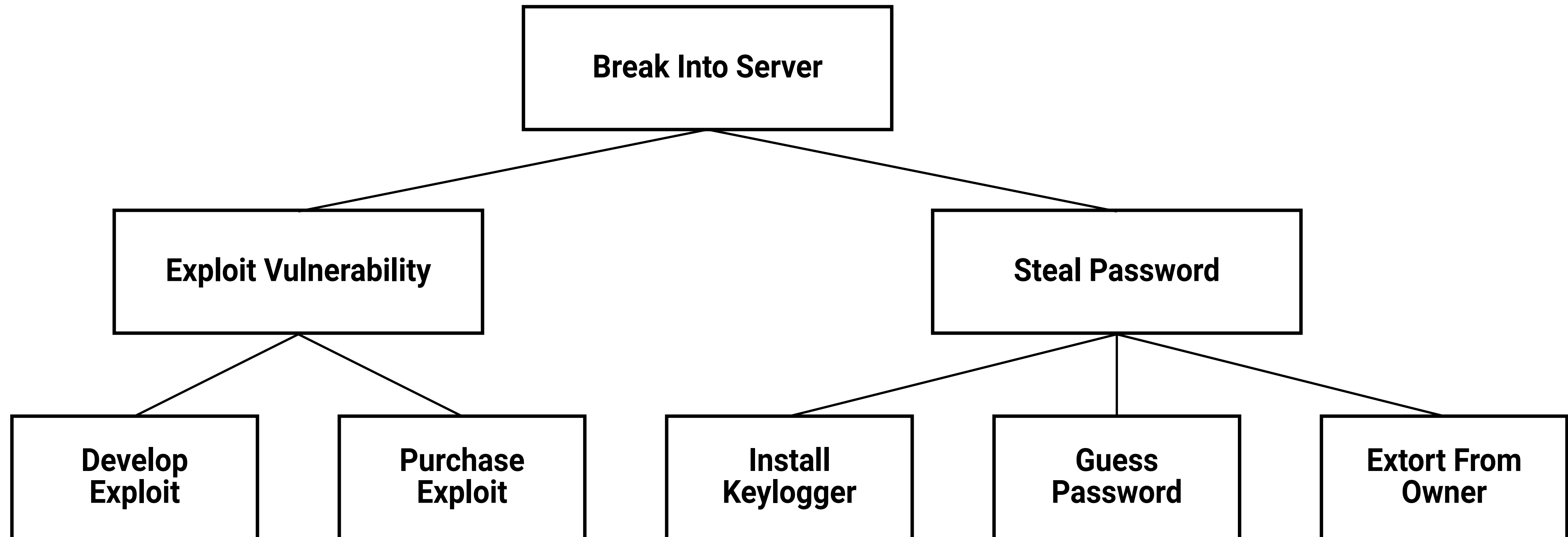
The paths an attacker may take

Impact of the attacks

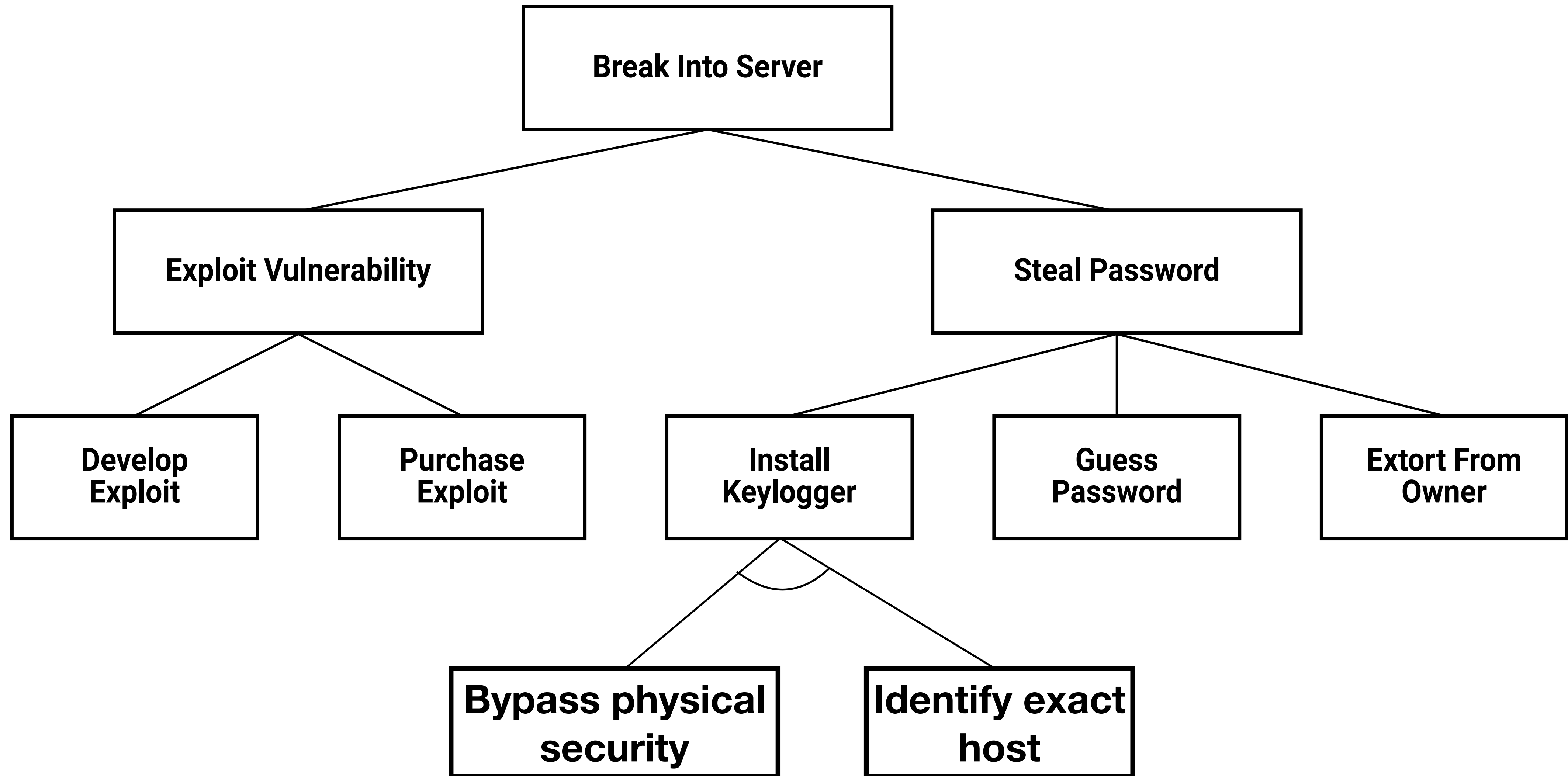
What countermeasures may mitigate against those attacks at particular points in the path

Prioritisation of countermeasures!

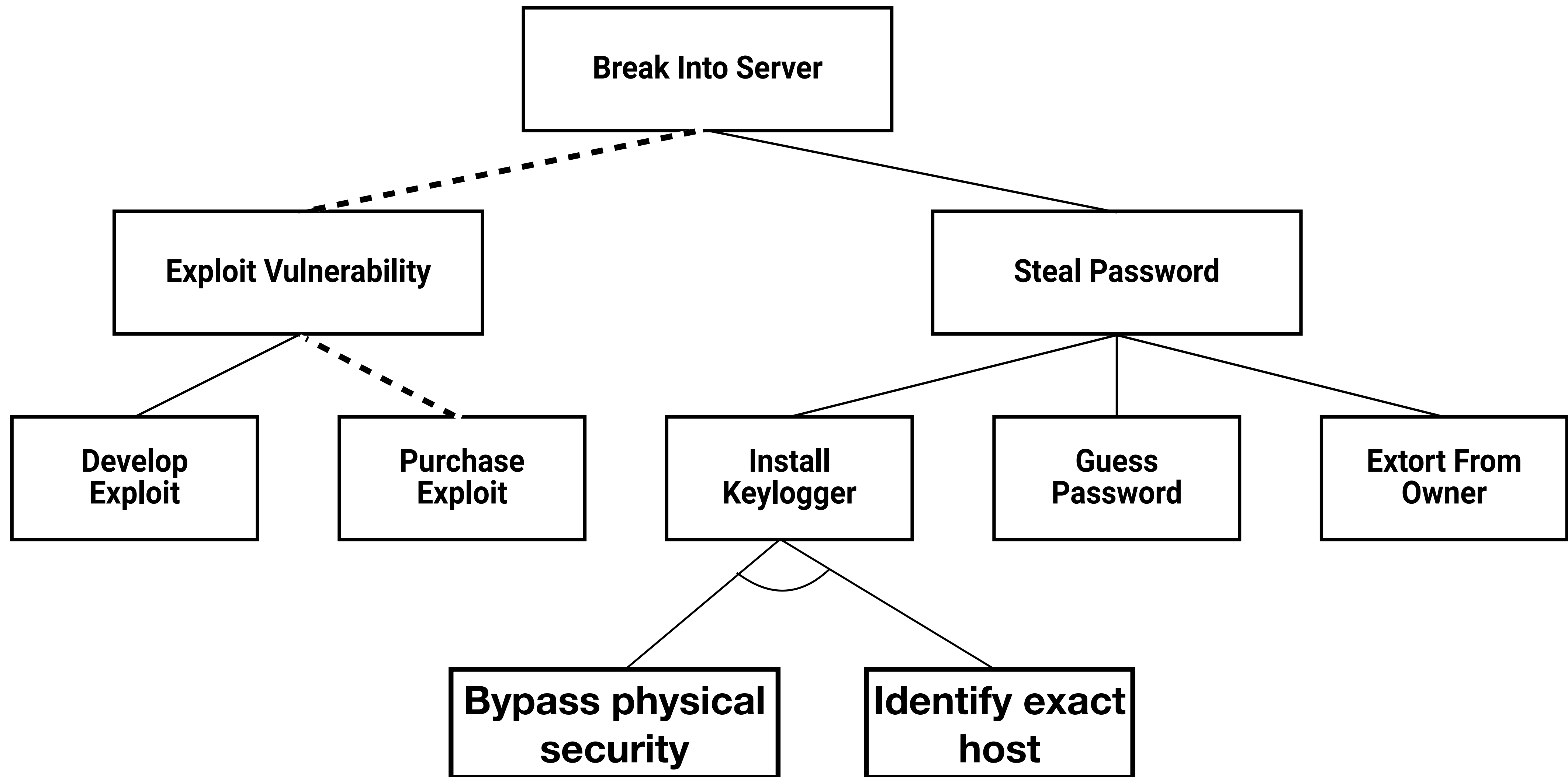
Attack trees



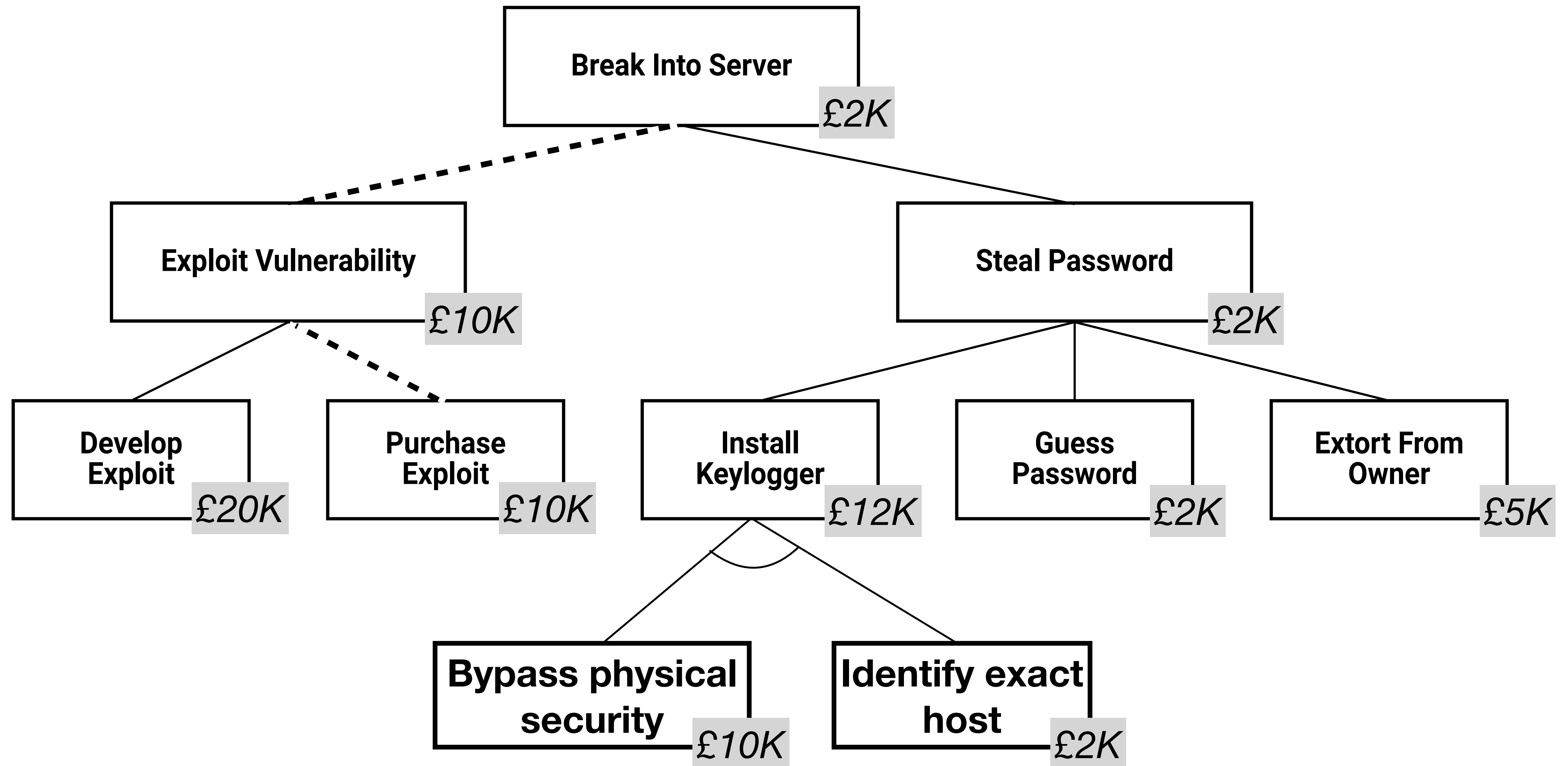
Attack trees - AND/OR combinations



Attack trees - possible/impossible paths



Attack trees - cost values

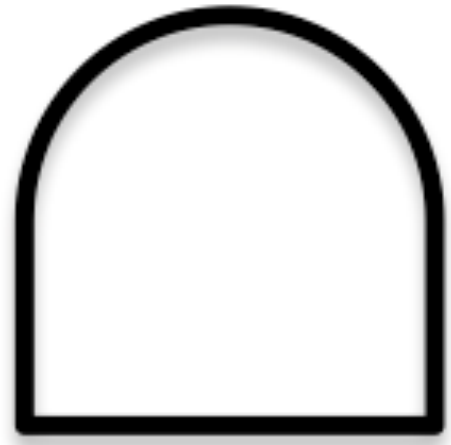


Limitations of Attack Trees

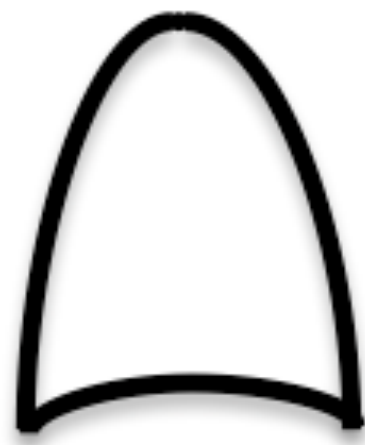
Single target. Alternative: *Attack Graphs* — Model attack actors, vectors, vulnerabilities and assets

Lack of rich semantics. Alternative: Incident Fault Trees

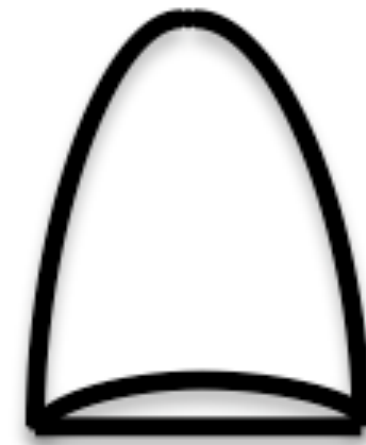
Incident Fault Trees



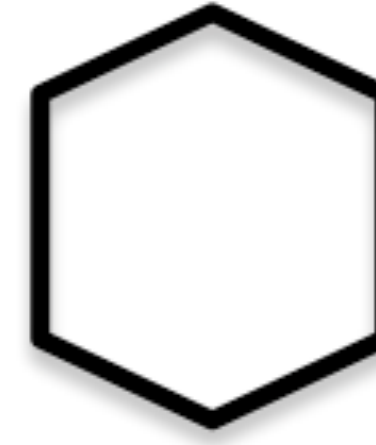
AND Gate



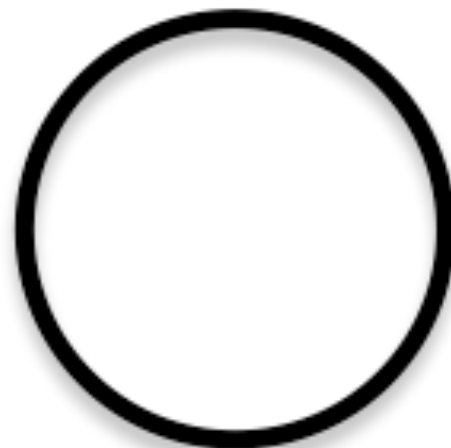
OR Gate



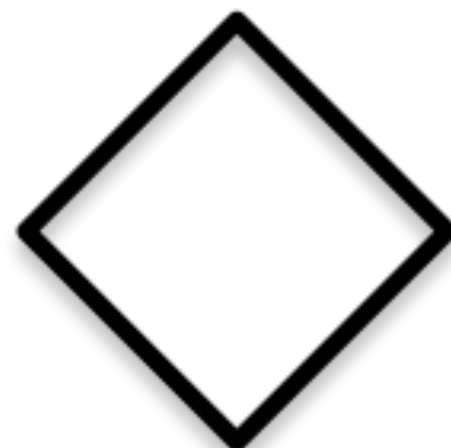
Exclusive
OR Gate



Inhibit
Gate



Basic
Event



Undeveloped
Event



Intermediate
Event



Conditioning
Event

Incident Fault Trees

Basic Event: cannot be explained in terms of other causal events, or has no known causes.

Intermediate Event: can be explained in terms of other causal events, or there are known causes for it.

Undeveloped Event: can be explained in terms of other causal events but we have chosen not to do so because such causal elaboration serves no analytical purpose.

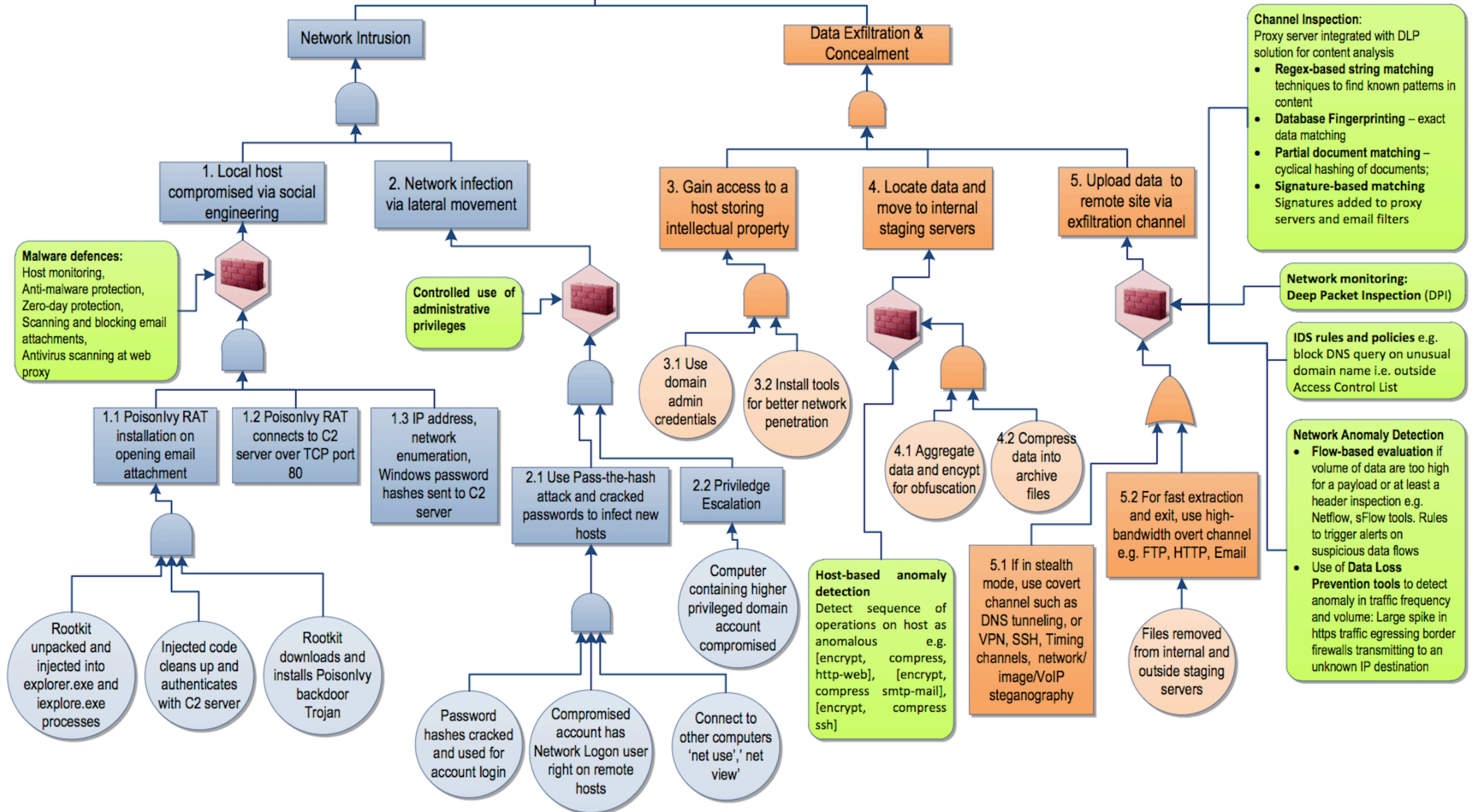
Incident Fault Trees

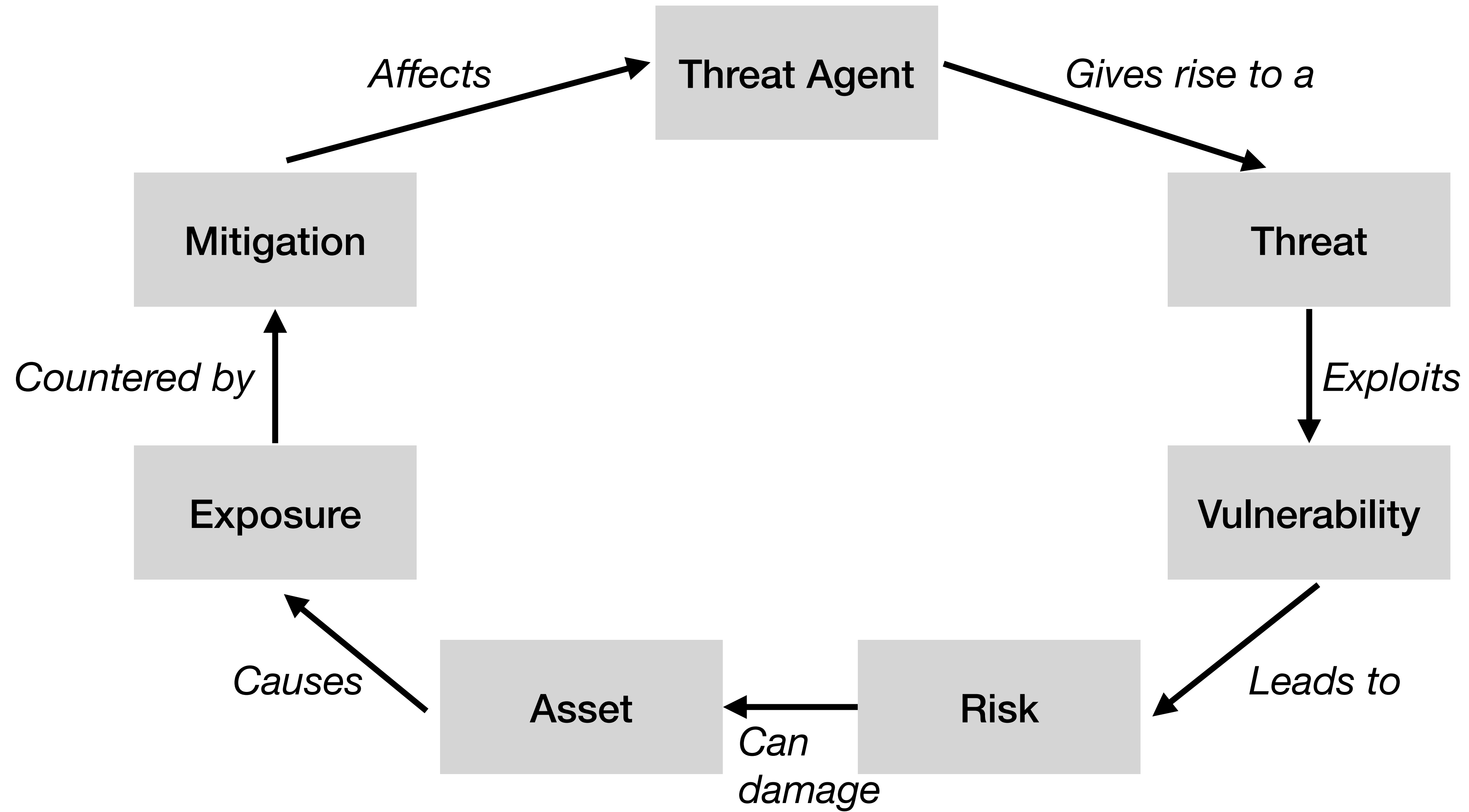
AND Gate and *OR Gate*: respectively represent the conjunction or disjunction of causal events.

Inhibit Gate: is placed between an event and its causes to stop an event from occurring unless the condition in the *Conditioning Event* is satisfied. We can use this combination to model mitigating or remedial actions that may act as barriers to the progress towards a security breach.

IFT for Nitro Attacks

Opportunistic Attack for IP Exfiltration





Further Reading

Listen to this podcast: <https://www.cybok.org/events/adversarial-behaviours-podcast>

Attack trees. Bruce Schneier. https://www.schneier.com/academic/archives/1999/12/attack_trees.html

Incident Fault Trees. *Rashid et al.* Discovering “Unknown Known” Security Requirements, Proc. International Conference on Software Engineering (ICSE), 2016. <https://doi.org/10.1145/2884781.2884785>